A two-feedback Vicsek-Couzin flock: density-stratified heading correlations as the scale-invariant fingerprint of motility–noise coupling

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Abstract

The standard Vicsek model holds the self-propulsion speed and the angular-noise distribution to global constants. Two strands of the active-matter literature have softened each in turn: density-dependent speed drives motility-induced phase separation, and heavy-tailed Lévy reorientations reshape the order–disorder transition. What happens when both adaptations operate together has not been resolved. We gate the speed and the stability index of an α-stable angular kick through one shared sigmoid in the local neighbour count, so crowded particles are simultaneously slow and Gaussian-like while isolated ones are fast and Cauchy-like. A controlled seven-size finite-size scan against the original Vicsek model and three single-channel ablations separates two regimes cleanly: only the motility-active modes develop a critical susceptibility scaling, and the full model is super-additive on that slope with respect to the motility-only ablation. The density-stratified heading correlation inside the dense phase, its dynamic counterpart $C(\tau)$, and the maximum-cluster size all read out the same mechanism — the noise-shape feedback rectifies Cauchy bursts into Gaussian-like kicks inside the motility-driven dense phase. A decoupled-sigmoid sweep further shows that the motility threshold alone controls the sign and amplitude of the rectification; the noise-shape threshold is essentially passive.

Highlights

- Two feedbacks — density-dependent motility and heavy-tailed Lévy reorientations — studied so far in isolation are activated together in the same Vicsek–Couzin flock through a shared sigmoid.
- Heavy-tailed reorientations on their own do not drive a critical $\chi_{\max}(L)$ scaling on the seven-size lever; only the motility channel does, and the noise-shape adaptation adds super-additively on top of it.
- The dense-quartile heading correlation records the rectification at $\Delta g \simeq +0.15$, $z \simeq +14$ at $L = 30$; its dynamic counterpart $C(\tau)$ holds the same gap over four decades in lag.
- A 25-cell decoupled-sigmoid sweep pins the motility threshold $n_\star^v$ as the sole control variable ($R^2 = 0.88$ on $n_\star^v$, $R^2 = 0.04$ on the noise-shape threshold). The density-triggered slowdown is a necessary precondition; without it, the heavy-tailed rectification damages coherence rather than building it.

1 Introduction

The standard Vicsek model [Vicsek et al., 1995] freezes two ingredients of a self-propelled flock that biology does not: every particle moves at the same speed v_0, and every angular kick is drawn from the same Gaussian. Both choices are physically wrong in interesting ways. Self-propulsion speed responds to local crowding in essentially every system one might want to model — confluent epithelia jam and arrest [Bi et al., 2016], swarming bacteria switch between planktonic and biofilm regimes through quorum sensing, and vehicular, pedestrian and animal traffic all carry a marked speed–density anti-correlation [Greenshields, 1935, Lighthill and Whitham, 1955]. The reorientation statistics are no closer to Gaussian: heavy-tailed Lévy turns are documented in foragers, T-cells and swarming bacteria [Reynolds, 2018, Harris et al., 2012, Ariel et al., 2015]. The biologically honest model lets both inputs depend on the local environment.

The two relaxations live in different parts of the literature. Cates and Tailleur [2015] proved that a density-dependent speed $v(\rho)$ with $\partial \ln v/\partial \ln \rho < -1$ drives motility-induced phase separation (MIPS) even with no alignment at all. Heavy-tailed angular noise has been studied on a separate track and is known to reshape the order–disorder transition of metric Vicsek dynamics. To our knowledge, the regime in which both adaptations are switched on together has not been resolved. The two channels are expected to interrogate each other inside the same dense regions: the noise distribution sets how particles escape spatial structures while the density-dependent speed sets those structures in the first place.

We tie the two adaptations together in the simplest way available. A single sigmoid in the local neighbour count n_i gates both the self-propulsion speed and the angular-noise stability index, so a crowded particle is simultaneously slow and Gaussian and an isolated one simultaneously fast and Cauchy. Switching both channels off ($v_i \equiv v_0$, $\alpha_i \equiv 2$) recovers the original Vicsek dynamics. Two single-channel ablations isolate the individual contributions: a motility-only mode (adaptive v_i, fixed Cauchy $\alpha = 1$) and a noise-shape-only mode (fixed $v_i = v_{\max}$, adaptive $\alpha_i \in [1, 2]$).

A short quantitative preview places the results that follow. The finite-size scan covers $L \in \{15, 22, 30, 45, 64, 90, 128\}$ at fixed density $\sigma = N/L^2 = 2.22$. The density-separation index s_{sep} plateaus at $\simeq 1.70$ at the largest sizes for both motility-active modes, against 1.13–1.16 for the Vicsek, Cauchy and noise-shape references: only the motility-active modes phase separate spatially. A log–log fit of $\chi_{\max} \sim L^a$ gives $a_{\mathrm{Vicsek}} = +0.01$, $a_{\mathrm{Cauchy}} = -0.12$, $a_{\mathrm{noise}} = +0.00$, $a_{\mathrm{motility}} = +2.34$ and $a_{\mathrm{full}} = +2.95$. The split between modes is qualitative, not quantitative: the three fixed-noise-shape modes carry slopes statistically indistinguishable from zero, with only the Cauchy reference robustly negative (95% bootstrap CI $[-0.15, -0.08]$); only the motility-active modes develop a critical $\chi_{\max}(L)$ scaling. The synergy diagnostic $\Delta = (a_{\mathrm{full}} - a_{\mathrm{Cauchy}}) - (a_{\mathrm{noise}} - a_{\mathrm{Cauchy}}) - (a_{\mathrm{motility}} - a_{\mathrm{Cauchy}})$ would vanish for two linearly-combining channels. It instead traces a persistently positive trajectory: $\Delta_4 = +0.69$, $\Delta_5 = +1.45$, $\Delta_6 = +0.46$, $\Delta_7 = +0.48$. Since the Cauchy slopes are essentially zero, Δ reduces in practice to $a_{\mathrm{full}} - a_{\mathrm{motility}} \simeq +0.6$.

The mechanism is direct: inside the dense phase the shared sigmoid pushes $\alpha_i \to 2$, and the Cauchy bursts that would otherwise scramble local alignment are rectified into Gaussian-like kicks. A ten-seed measurement at $L = 30$ gives $g(r \simeq 0.6) = 0.542 \pm 0.007$ for the motility-only ablation against 0.695 ± 0.008 for the double-adaptive model, a paired gap of $+0.153 \pm 0.011$ ($z = +13.6$); the dilute-quartile correlation is essentially flat between the two modes. The cluster geometry confirms the picture: the maximum-cluster size grows from 55.0 ± 7.5 particles in motility-only to 176.6 ± 42.3 in the double-adaptive model ($z = +2.7$) at conserved total dense mass. A decoupled-sigmoid sweep (Sec. 3.3.4) pins the control variable: on a 25-cell grid the dense-quartile gap follows $\Delta g \simeq -0.144\, n_\star^v + 0.52$ with $R^2 = 0.88$, while the analogous fit on $n_\star^\alpha$ captures only $R^2 = 0.04$. Biologically, the density-triggered slowdown is a necessary precondition; without it, the heavy-tailed rectification damages dense-phase coherence instead of building it.

2 Model and methods

2.1 Dynamics

N point particles live in a square box of side L with periodic boundary conditions; particle i carries a position $\mathbf{x}_i \in [0, L)^2$ and a heading $\theta_i \in (-\pi, \pi]$, advanced synchronously through

$$\theta_i(t+1) = \theta_i^\star(t+1) + \xi_i(t), \qquad \mathbf{x}_i(t+1) = \mathbf{x}_i(t) + v_i(t+1)\,\vec{e}_i(t+1) \pmod{L}, \tag{1}$$

with $\vec{e}_i = (\cos\theta_i, \sin\theta_i)$. The desired heading $\theta_i^\star$ follows the Couzin priority rule [Couzin et al., 2002] with the repulsion set $\mathcal{R}_i = \{j \neq i : |\mathbf{r}_{ij}| < R_r\}$, $\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i$, and the alignment set $\mathcal{A}_i = \{j \neq i : R_r \leq |\mathbf{r}_{ij}| < R_a\}$, both sensed omnidirectionally. Repulsion takes priority over alignment, alignment over inertia:

$$\theta_i^\star(t+1) = \begin{cases} \arg\Big(-\sum_{j\in\mathcal{R}_i} \mathbf{r}_{ij}/|\mathbf{r}_{ij}|\Big), & \mathcal{R}_i \neq \emptyset, \\ \arg\Big(\sum_{j\in\mathcal{A}_i} e^{i\theta_j(t)}\Big), & \mathcal{R}_i = \emptyset \text{ and } \mathcal{A}_i \neq \emptyset, \\ \theta_i(t), & \mathcal{R}_i = \mathcal{A}_i = \emptyset. \end{cases} \tag{2}$$

Interactions run in $O(N)$ on a cell-list partition.

The angular kick ξ_i is a symmetric α-stable variate with a particle-dependent stability index,

$$\xi_i(t) \sim S_{\alpha_i(t)}\big(c = \eta,\, \beta_{\text{stab}} = 0,\, \mu = 0\big), \tag{3}$$

sampled with the Chambers–Mallows–Stuck representation [Chambers et al., 1976]. The Gaussian limit $\alpha_i = 2$ is handled separately and recovers the original Vicsek noise.

2.2 The shared sigmoid

The single departure from canonical Vicsek dynamics is that both the speed and the noise stability index become local fields, controlled by the same neighbour count $n_i(t) = |\mathcal{A}_i(t)|$ through one sigmoid,

$$\sigma_i(t) = \frac{1}{1 + \exp\big[-s\,(n_i(t) - n_\star)\big]}, \tag{4}$$

$$v_i(t) = v_{\max} - (v_{\max} - v_{\min})\,\sigma_i(t), \tag{5}$$

$$\alpha_i(t) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\,\sigma_i(t). \tag{6}$$

A crowded particle ($n_i \gg n_\star$) sits at $\sigma_i \to 1$, $v_i \to v_{\min}$, $\alpha_i \to \alpha_{\max} = 2$; an isolated one ($n_i \ll n_\star$) at $\sigma_i \to 0$, $v_i \to v_{\max}$, $\alpha_i \to \alpha_{\min} = 1$. The two channels are locked: slow means Gaussian, fast means Cauchy.

2.3 Reference and ablation modes

The two-feedback model (*double-adaptive*) is benchmarked against three controls on the same code path. Setting $v_{\min} = v_{\max}$ and $\alpha_{\min} = \alpha_{\max} = 2$ recovers the original Vicsek dynamics (*Vicsek reference*); setting instead $\alpha_{\min} = \alpha_{\max} = 1$ gives the heavy-tailed *Cauchy reference*, the backdrop on top of which the two feedbacks are switched on. The *noise-shape only* mode activates the noise-shape channel with v_i frozen at $v_{\max}$; the *motility only* mode activates the motility channel with α_i frozen at the Cauchy value.

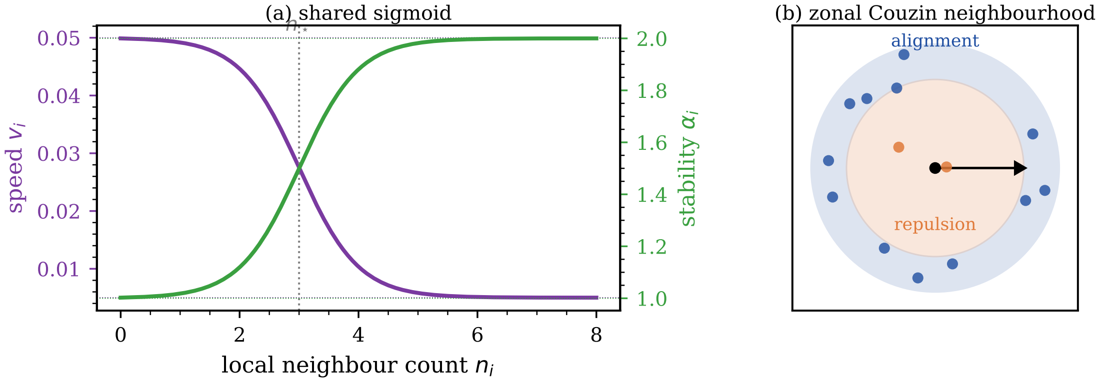

Figure 1. The two-feedback model. (a) The shared sigmoid maps n_i onto the propulsion speed v_i (purple) and the stability index α_i (green), which move together as the particle goes from isolated to crowded. (b) The Couzin zonal neighbourhood: inner repulsion disc and outer alignment annulus, sensed omnidirectionally.

2.4 Order parameters and observables

The polar order parameter is the magnitude of the average heading,

$$\langle\varphi\rangle = \left\langle \frac{1}{N}\Big|\sum_i e^{i\theta_i}\Big| \right\rangle, \tag{7}$$

the susceptibility its scaled variance $\chi = N\,\mathrm{Var}(\varphi)$, and the Binder cumulant $U_4 = 1 - \langle\varphi^4\rangle/(3\langle\varphi^2\rangle^2)$. For spatial phase separation we use

$$s_{\mathrm{sep}} \equiv \max_{\mathrm{bins}} \frac{n_{\mathrm{bin}}}{\mathrm{med}(n_{\mathrm{bin}})}, \tag{8}$$

the ratio of the densest grid cell to the median cell on a 10×10 partition: a homogeneous fluid sits at $s_{\mathrm{sep}} \simeq 1$, a phase-separated configuration well above. A grid-free pair-correlation analogue (the 95th-percentile / median ratio of the per-particle local density within radius $R = 1$) ranks the modes consistently and ships with the code. To distinguish a traveling-band soliton from an isotropic patch we use the band-soliton index

$$b \equiv \frac{\max_{x_\parallel} \sigma(x_\parallel)}{\langle\sigma\rangle}, \tag{9}$$

the maximum of the time-averaged density profile along the instantaneous polar-flow axis $x_\parallel$ divided by the spatial mean. A metric-Vicsek traveling band gives $b \gtrsim 1.5$; an isotropic patch gives $b \simeq 1$.

2.5 Numerical protocol

The seven sizes $L \in \{15, 22, 30, 45, 64, 90, 128\}$ run at fixed density $\sigma = N/L^2 = 2.22$, three seeds per (mode, L, η) cell, with 1500 warm-up and 1000 measurement steps from $\theta_i \equiv 0$. The noise grid is $\eta \in \{0.005, 0.010, 0.020, 0.035, 0.050, 0.075, 0.100, 0.150, 0.200, 0.300\}$. Fixed parameters: $R_r = 0.5$, $R_a = 0.7$, $n_\star = 3$, $s = 2$, $v_{\min} = 0.005$, $v_{\max} = 0.05$, $\alpha_{\min} = 1$, $\alpha_{\max} = 2$. The interaction loop is numba-JIT compiled. A small-η refinement $\{0.001, 0.002, 0.003, 0.0075\}$ at the first five sizes (two seeds) confirms that $\chi_{\max}$ does not lurk outside the original window, except for the noise-shape-only ablation at $L = 45$, which we report with its updated value.

3 Results

We first set a visual reference. The double-adaptive model at $L = 30$, $N = 2000$, after 3000 warm-up steps, at three noise levels spanning the ordered, near-critical and disordered regimes

(Fig. 2): the flock is almost fully aligned and dense at $\eta = 0.020$; an inhomogeneous configuration of locally ordered slow patches in a faster disordered gas at near-critical $\eta = 0.10$; particles fill the box without a coherent global heading at $\eta = 0.30$. The near-critical panel is the operating point at which the rectification mechanism documented below is active.

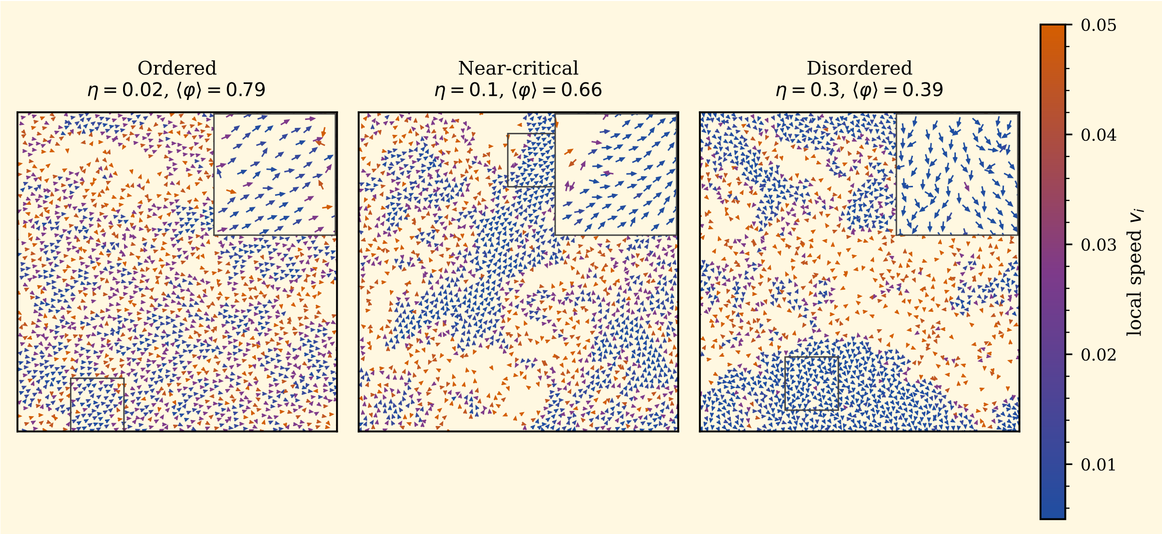

Figure 2. Real-space configurations of the double-adaptive model at $L = 30$, $N = 2000$, single seed, after 3000 warm-up steps, at three noise levels spanning the ordered ($\eta = 0.020$), near-critical ($\eta = 0.10$) and disordered ($\eta = 0.30$) regimes on a cream background. Each particle is drawn as an arrow whose colour encodes the local speed v_i on a navy–mauve–vermillion colormap (shared colorbar on the right); a 5×5 zoom inset on the densest sub-region accompanies each panel. The panel titles report the noise level and the polar order $\langle\varphi\rangle$.

3.1 Global diagnostics

3.1.1 Finite-size scaling of the susceptibility

To probe whether the two adaptations leave a thermodynamic trace we run all four heavy-tailed modes on the seven-size grid of Sec. II E and read the susceptibility peak $\chi_{\max}(L)$; its slope a in $\chi_{\max} \sim L^a$ separates the four modes (Fig. 3, Table 1).

The slopes split cleanly into two regimes. The three fixed-noise-shape modes (Vicsek–Gaussian, Cauchy, noise-shape only) carry slopes small in absolute value ($+0.01$, -0.12, $+0.00$); the Vicsek and noise-shape intervals straddle zero ($[-0.09, +0.06]$ and $[-0.18, +0.20]$), while the Cauchy reference is robustly negative ($[-0.15, -0.08]$), with its $\chi_{\max}$ in fact *dropping* from 1.32 at $L = 15$ to 0.98 at $L = 128$. None develops a critical scaling on the seven-point lever. The two motility-active modes scale strongly: $a_{\text{motility}} = +2.34$, $a_{\text{full}} = +2.95$, the full mode robustly steeper. The synergy diagnostic

$$\Delta \equiv (a_{\text{full}} - a_{\text{Cauchy}}) - (a_{\text{noise}} - a_{\text{Cauchy}}) - (a_{\text{motility}} - a_{\text{Cauchy}}) \tag{10}$$

vanishes for two linearly-combining channels above the Cauchy backdrop. With our Cauchy and noise-shape slopes both essentially zero, Δ reduces to $a_{\text{full}} - a_{\text{motility}}$ and the seven-size fit returns $\Delta_7 = +0.48$: the noise-shape rectification, which carries no critical exponent on its own, adds an excess slope of order $+0.5$ on top of the motility-driven scaling.
The cumulative size budget is consistent. The four smallest sizes give $\Delta_4 = +0.69$ ($[+0.25, +1.19]$, 95%); five sizes give $\Delta_5 = +1.45$ ($[+0.60, +2.44]$); six give $\Delta_6 = +0.46$ and seven give $\Delta_7 = +0.48$. The synergy is significant at $n = 4, 5$ and persists at roughly half that magnitude through $L = 128$. The interval widens at $n = 6, 7$ because the motility slope itself carries a wide bootstrap dispersion ($[+1.60, +3.04]$), but the point estimate stays positive.

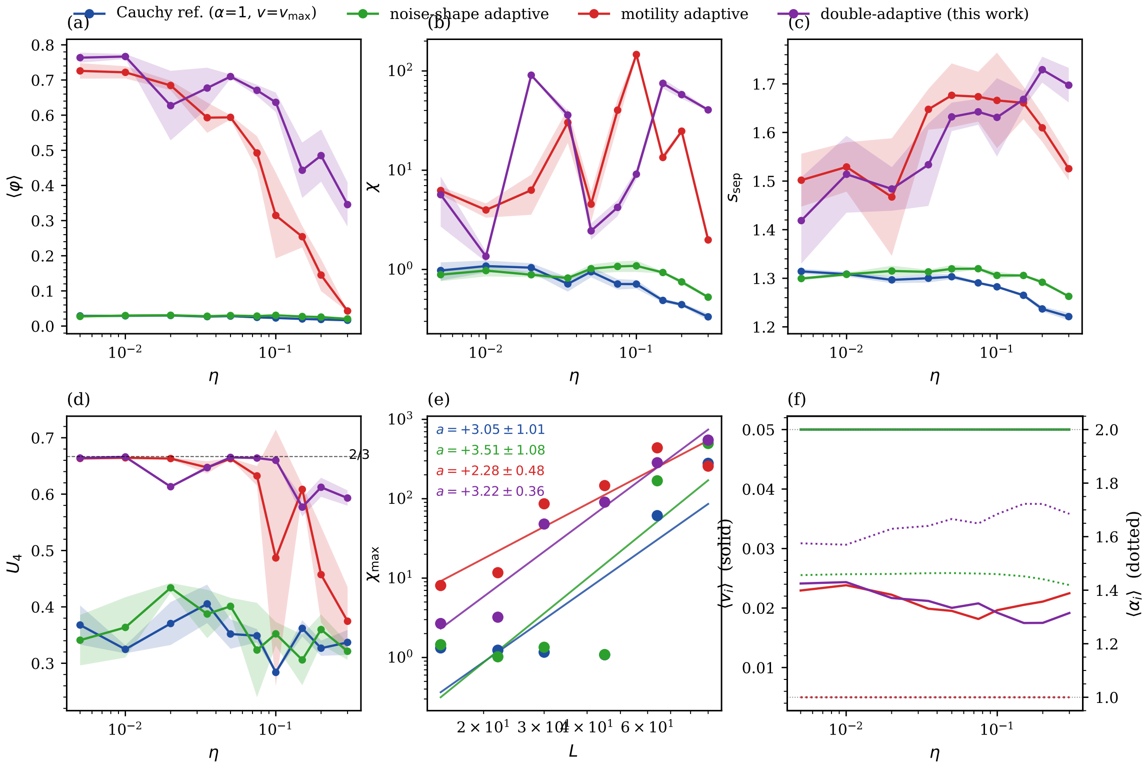

Figure 3. Seven-size scan, $L \in \{15, 22, 30, 45, 64, 90, 128\}$, three seeds, four study modes: heavy-tailed Cauchy reference (blue, `#1f4ea1`), noise-shape only (green, `#2ca02c`), motility only (red, `#d62728`), double-adaptive (purple, `#7e2aa0`). Panel (a) shows the polar order $\langle \varphi \rangle$ versus η at the largest L; panel (b), the susceptibility χ; panel (c), the density-separation index s_{sep}; panel (d), the Binder cumulant U_4. Panel (e) is the log–log $\chi_{\max}$ versus L with fitted slopes per mode; panel (f) reports the population mean speed $\langle v_i \rangle$ (solid) and the population mean stability $\langle \alpha_i \rangle$ (dotted) versus η at the largest L.

The reading is mechanistic. Inside a dense patch the alignment sum over the many neighbours in the annulus is dominated by typical kicks rather than by rare extreme ones, so the Cauchy bursts of the heavy-tail-only modes are averaged out before they can drive a $\chi_{\max}$ growth. The motility channel sidesteps this averaging: the dense phase it carves out has a susceptibility set by density–density fluctuations rather than by single-particle bursts. The noise-shape rectification adds on top because, in the same dense patch, $\alpha_i \to 2$ damps the residual heading fluctuations, producing the extra $\Delta a \simeq +0.6$ between the full and motility-only slopes.

The U_4 panel in Fig. 3d adds a caveat. All four modes, including the Cauchy reference, reach the same $U_4 \to 0.28$–0.31 dip depth at $L = 64$. Cauchy fluctuations of the global order in the dilute phase push U_4 down on their own without any two-phase coexistence: the dip is not a reliable first-order signature in heavy-tailed Vicsek dynamics. We turn to s_{sep}.

3.1.2 Density-separation index

The next question is whether the dense regions are spatially separated. Computing s_{sep} on the same protocol gives, at the χ-peak η and the three largest sizes, $s_{\text{sep}} = (1.25, 1.25, 1.48, 1.72)$ at $L = 64$, $(1.19, 1.18, 1.68, 1.78)$ at $L = 90$, and $(1.13, 1.14, 1.69, 1.70)$ at $L = 128$, in the order Cauchy, noise-shape, motility, full (Fig. 3c). The two Cauchy-only modes are essentially homogeneous ($s_{\text{sep}} \simeq 1.2$, the binning floor) and their s_{sep} in fact decreases slowly with L. Only the two motility-active modes phase separate, both plateauing near $s_{\text{sep}} \simeq 1.7$. The double-adaptive model sits above the motility-only ablation by $+0.24$ at $L = 64$, $+0.10$ at $L = 90$, and $+0.01$ at $L = 128$, the two modes converging on the asymptotic plateau.

Table 1. Susceptibility peaks $\chi_{\max}(L)$ on the seven-size grid and log–log FSS slopes a for the Vicsek–Gaussian reference and the four heavy-tailed modes, three seeds. The 95% bootstrap CI on the slope ($B = 10\,000$ resamples of the seven $(L, \chi_{\max})$ pairs) is wide because of the seven-point fitting budget. The seed-level $\chi_{\max}$ standard deviation is typically 20–60% of the cell value at the four smallest sizes (per-seed channels are not stored at $L \geq 64$). The three fixed-noise-shape modes stay flat within ± 0.4 across the full size range, an order of magnitude below the two motility-active modes.

Mode	$L = 15$	$L = 22$	$L = 30$	$L = 45$	$L = 64$	$L = 90$	$L = 128$	slope a	95% CI
Vicsek (Gaussian)	1.14	1.38	1.20	1.17	1.18	1.27	1.24	+0.01	[−0.09, +0.06]
Cauchy reference	1.32	1.24	1.17	1.08	1.11	1.07	0.98	−0.12	[−0.15, −0.08]
noise-shape adaptive	1.45	1.02	1.35	1.09	1.12	1.20	1.42	+0.00	[−0.18, +0.20]
motility adaptive	8.07	11.70	86.34	146.44	189.56	1119.24	736.53	+2.34	[+1.60, +3.04]
double-adaptive	2.68	3.22	47.99	90.62	583.94	293.99	1079.14	+2.95	[+1.93, +4.13]

3.1.3 Direct visualisation in real space

The next question is how motility-only and double-adaptive differ on individual configurations across the order–disorder window. We run both at $L = 30$, $N = 2000$, single seed, 3000 warm-up steps, at three noise levels (Fig. 4).

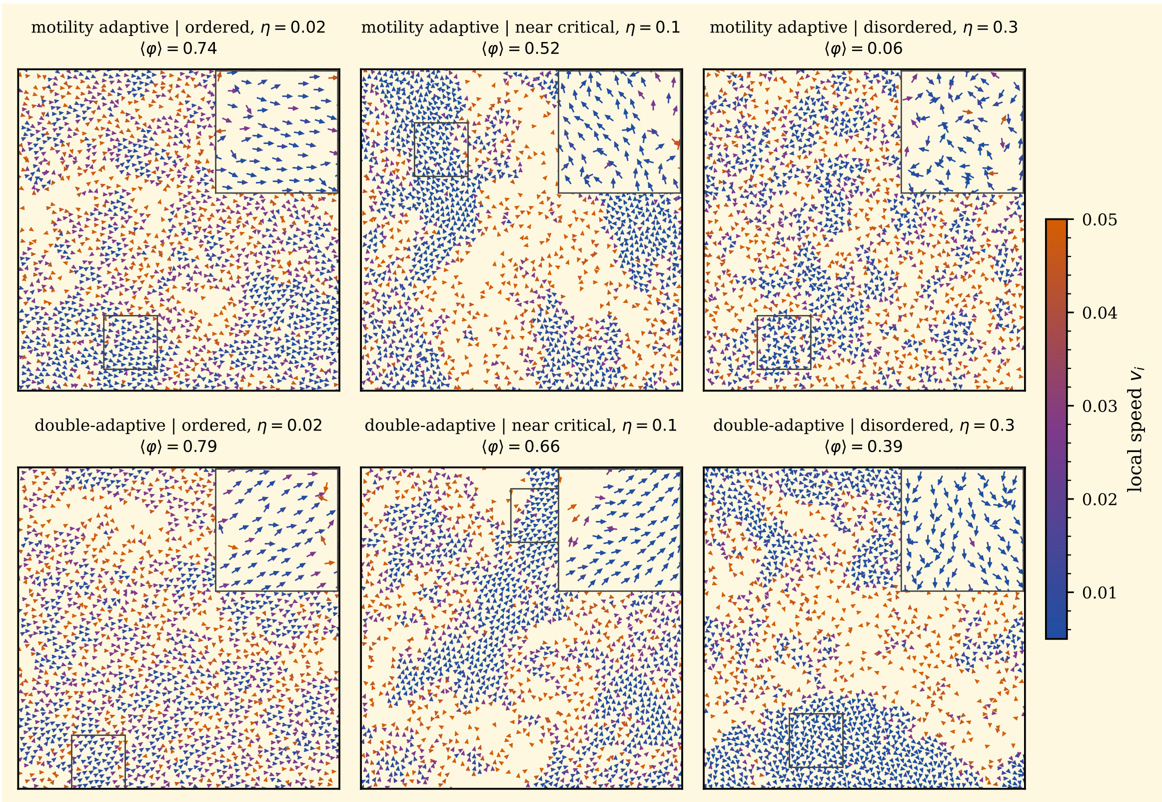

Figure 4. Real-space configurations at $L = 30$, $N = 2000$, after 3000 warm-up steps, single seed, on a cream background. Top row: motility only. Bottom row: double-adaptive. Columns left to right: $\eta = 0.020$ (deeply ordered), $\eta = 0.10$ (near-critical), $\eta = 0.30$ (deeply disordered). Each particle is drawn as an arrow coloured by its instantaneous local speed v_i on a navy–mauve–vermillion colormap (shared colorbar on the right); a 5×5 zoom inset on the densest sub-region accompanies each panel.

Deep in the ordered phase at $\eta = 0.020$ the two modes are almost indistinguishable, with $\langle\varphi\rangle = 0.78$ and 0.83 respectively. At near-critical $\eta = 0.10$ the double-adaptive model already holds more order ($\langle\varphi\rangle$ rises from 0.47 to 0.63). The contrast becomes sharp at $\eta = 0.30$: the motility ablation collapses to $\langle\varphi\rangle = 0.04$, while the double-adaptive model still holds 0.25. The dense slow patches at the bottom right of the figure remain coherently aligned; the same patches in the top row are scrambled. Inside the dense phase $\langle \alpha_i \rangle$ reaches 1.5–1.6 for the full mode and damps the Lévy bursts that would otherwise reshuffle the slow particles; the motility-only ablation takes the full Cauchy kick at every step in those same regions. The visual signature matches the shared sigmoid: rectification acts in the dense regions, and nowhere else.

3.2 Parametric analyses and distributions

3.2.1 Behavioural plane in $(n_\star, s)$

The single working point used so far ($n_\star = 3$, $s = 2$) sets the strength of both feedbacks. To check whether the contrast extends to the whole plane we sweep $n_\star$ and s on a 5×4 grid at $L = 22$, two seeds, with a five-point η sub-grid resolving the peaks (Fig. 5).

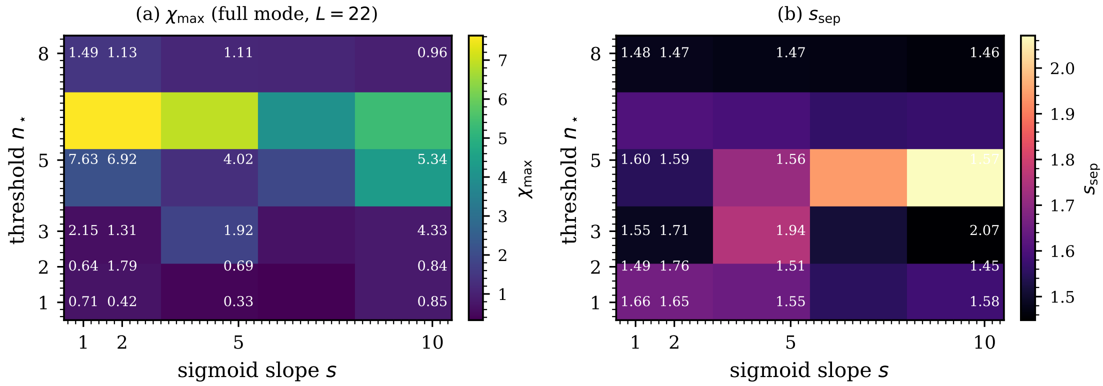

Figure 5. Behavioural-parameter plane of the double-adaptive model at $L = 22$, two seeds, five-point eta sub-grid. Panel (a) shows the susceptibility peak $\chi_{\max}$, panel (b) the density-separation index $s_{\rm sep}$. Cell labels report numerical values.

In the upper rectangle $n_\star \geq 5$ the index falls back to $s_{\rm sep} \simeq 1.50$: the threshold sits past the typical occupancy $\bar{n}_i \simeq \sigma\pi(R_a^2 - R_r^2) \simeq 1.7$ and neither feedback engages. In the lower rectangle $n_\star \leq 3$ with $s \geq 5$ the index climbs to $s_{\rm sep} = 1.84$–1.87 and peaks at $(n_\star, s) = (3, 10)$. The cohesion gain covers the entire MIPS-active corner rather than sitting on a knife-edge cell. The susceptibility map in panel (a) is noisier at two seeds, so we re-sweep the same plane at $L = 30$ with five seeds.

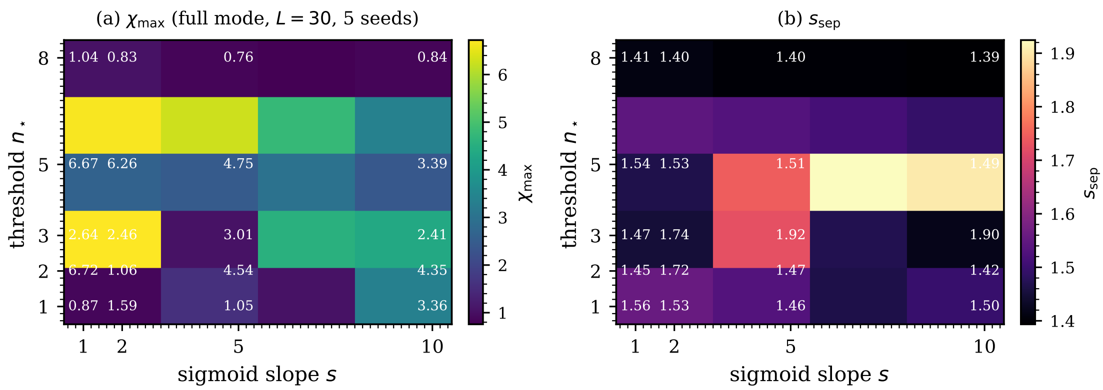

Figure 6. Refined behavioural plane at $L = 30$ with five seeds. Panel (a) reports the susceptibility peak $\chi_{\max}$, panel (b) the density-separation index $s_{\rm sep}$.

The refined density-separation panel preserves the $L = 22$ structure: $n_\star \geq 5$ falls back to $s_{\rm sep} \leq 1.49$, while $n_\star \leq 3$ with $s \geq 5$ reaches 1.68–1.76 and peaks at $(2, 10)$. The susceptibility map no longer co-localises. The brightest cell moves down to the $n_\star = 1$ row and peaks at

$(1, 10)$ with $\chi_{\max} = 31.9$; the whole $n_\star \le 2$ rectangle is brighter than the $n_\star = 3$ row, and the $\chi_{\max}$ peak sits where $s_{\rm sep}$ has dropped to 1.48. Cohesion and susceptibility therefore optimise on different cells: the susceptibility peaks where the threshold sits below typical occupancy and the rectification fires almost everywhere (washing out the dense–dilute asymmetry), while $s_{\rm sep}$ peaks where the threshold bridges typical occupancy and the rectification stays selective for the dense phase.

3.2.2 Fine-L scan of the transient advantage

We add five intermediate sizes $L \in \{38, 50, 60, 75, 105\}$ at five seeds each with a four-point η sub-grid to track the $s_{\rm sep}$ gap between the seven FSS nodes (Fig. 7).

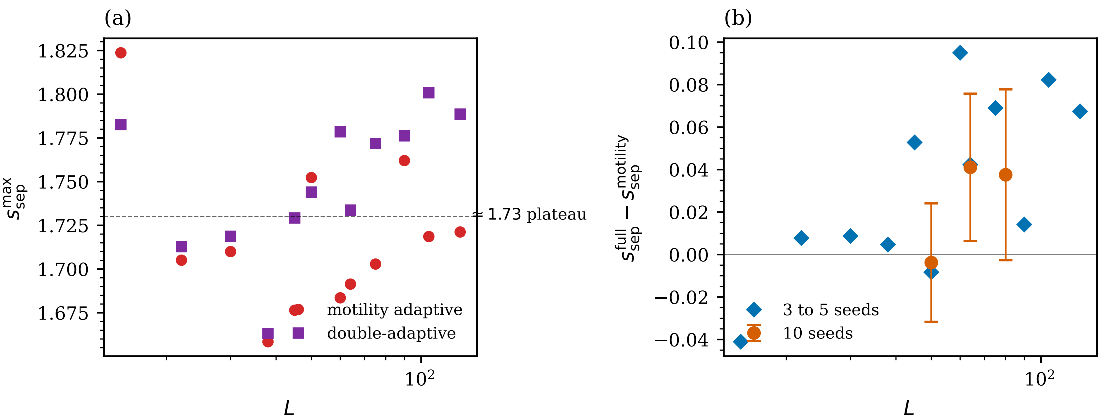

Figure 7. Twelve-size $s_{\rm sep}$ scan combining the seven-size controlled FSS (three seeds) with five intermediate sizes $L \in \{38, 50, 60, 75, 105\}$ (five seeds). Panel (a): $s_{\rm sep}^{\max}$ vs L for motility-only and double-adaptive. Panel (b): the gap $s_{\rm sep}^{\rm full}(L) - s_{\rm sep}^{\rm motility}(L)$.

The gap is small at $L = 38, 50$ ($\le |0.01|$) and turns positive at $L = 60, 75, 105$ ($+0.10, +0.07, +0.08$). The closure at the two smaller fine-grid sizes follows the mode-dependent location of the $s_{\rm sep}$ peak in η. A ten-seed paired gap at the full-mode χ-peak η for $L \in \{50, 64, 80\}$ (Table 2) gives $+0.020 \pm 0.032$, $+0.165 \pm 0.031$, $+0.024 \pm 0.031$ respectively: only $L = 64$ is robustly positive.

Table 2. Paired $s_{\rm sep}$ gap between double-adaptive and motility-only at the full-mode χ-peak η, ten seeds per cell.

L	gap (full − motility)	SE	z
50	+0.020	0.032	+0.64
64	+0.165	0.031	+5.28
80	+0.024	0.031	+0.77

The $s_{\rm sep}$ advantage of the double-adaptive model is therefore real and significant at $L \simeq 64$ but small and seed-budget-limited at neighbouring sizes. The susceptibility synergy carries the cleanest signature of the noise-shape contribution; $s_{\rm sep}$ corroborates it in the intermediate-L window.

3.2.3 Order-parameter distribution at $L = 30$

The Binder dip is ambiguous on its own: a weakly first-order transition deepens U_4 through a bimodal $P(\langle\varphi\rangle)$, while a continuous transition with large γ/ν deepens it without bimodality. Only the full distribution separates the two cases. We histogram $\langle\varphi\rangle$ on long trajectories at $L = 30$, $N = 2000$, three seeds, 6000 measurement steps each, for the motility-only ablation at $\eta = 0.10$ and the double-adaptive model at $\eta = 0.15$ (Fig. 8). Both histograms come out unimodal.

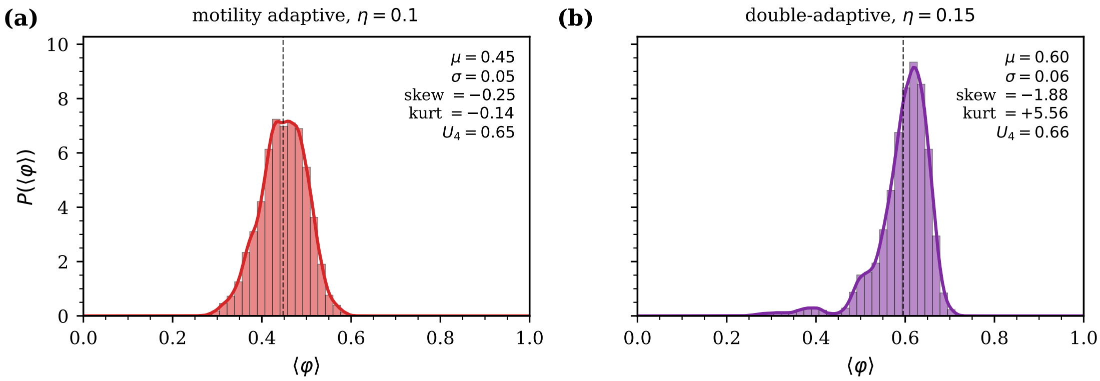

Figure 8. Histograms of $\langle\varphi\rangle$ at $L = 30$, $N = 2000$, three seeds, 6000 measurement steps each. Panel (a): motility-only ablation at the near-critical $\eta = 0.10$. Panel (b): double-adaptive model at $\eta = 0.15$.

Motility-only peaks at $\langle\varphi\rangle \simeq 0.45$ with skewness -0.25, kurtosis -0.14 (a mildly left-skewed broad mode); the double-adaptive model peaks at 0.60 with a sharper core (kurtosis $+5.56$) and a long left tail (skewness -1.88). The long-trajectory Binder cumulants $U_4 = 0.65$ (motility) and 0.66 (full) sit just below the bimodal limit $U_4 \to 2/3$ and well above the Gaussian $U_4 = 0$: the trajectories are locked in a single ordered state with bounded fluctuations, not switching between two phases. The transition is at most weakly first-order at this size.

Rerunning the histograms at $L = 64, 90$ (five seeds, 6 000 steps) and at $L = 128$ (five seeds, 60 000 steps), the double-adaptive distribution sits to the right of motility-only at every size: $\Delta\langle\varphi\rangle = +0.145, +0.048, +0.121$ at $L = 64, 90, 128$ (the $L = 90$ row is a seed-budget outlier already seen in s_{sep}). The full-mode U_4 stays in $[0.57, 0.62]$ and the distributions broaden into mildly platykurtic shapes at $L = 128$ ($\kappa = -0.27$ for motility, -0.20 for full) without developing two-peak structure. $P(\langle\varphi\rangle)$ remains unimodal across all four sizes.

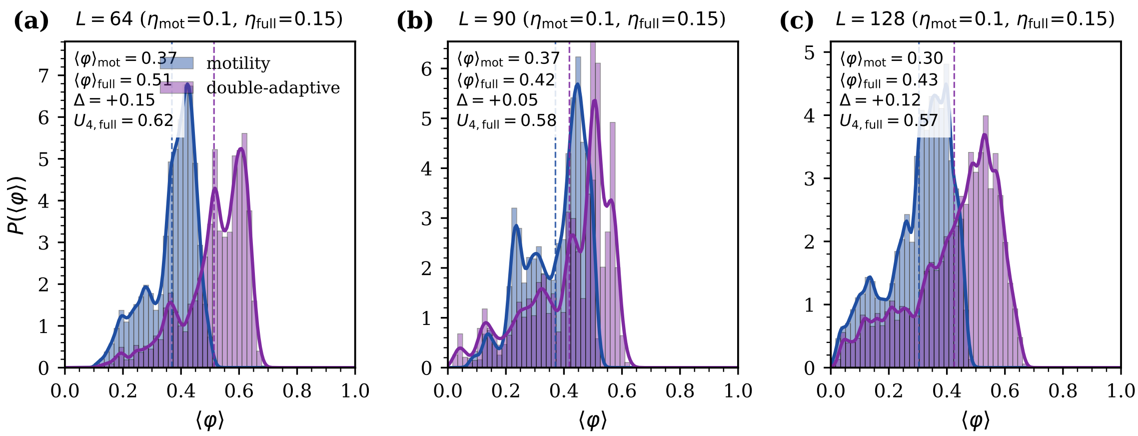

Figure 9. Histograms of $\langle\varphi\rangle$ at $L = 64, 90$ (five seeds, 6 000 steps) and at $L = 128$ (five seeds, 60 000 steps), motility-only at $\eta = 0.10$ and double-adaptive at $\eta = 0.15$. Vertical dashed lines mark the per-mode means; gaps are $+0.145, +0.048, +0.121$ at $L = 64, 90, 128$ ($L = 90$ is a seed-budget outlier).

3.2.4 Time-averaged density profile along the polar axis

The dense phase still has to be characterised geometrically: does it travel as a metric-Vicsek soliton or sit as an isotropic patch comoving with the flow? The band-soliton index b of Eq. (9) discriminates between the two ($b \gtrsim 1.5$ for a band, $b \simeq 1$ for a patch). We measure b for all four heavy-tailed modes at their near-critical η, $L = 30$, three seeds, 4000 measurement steps (Fig. 10).

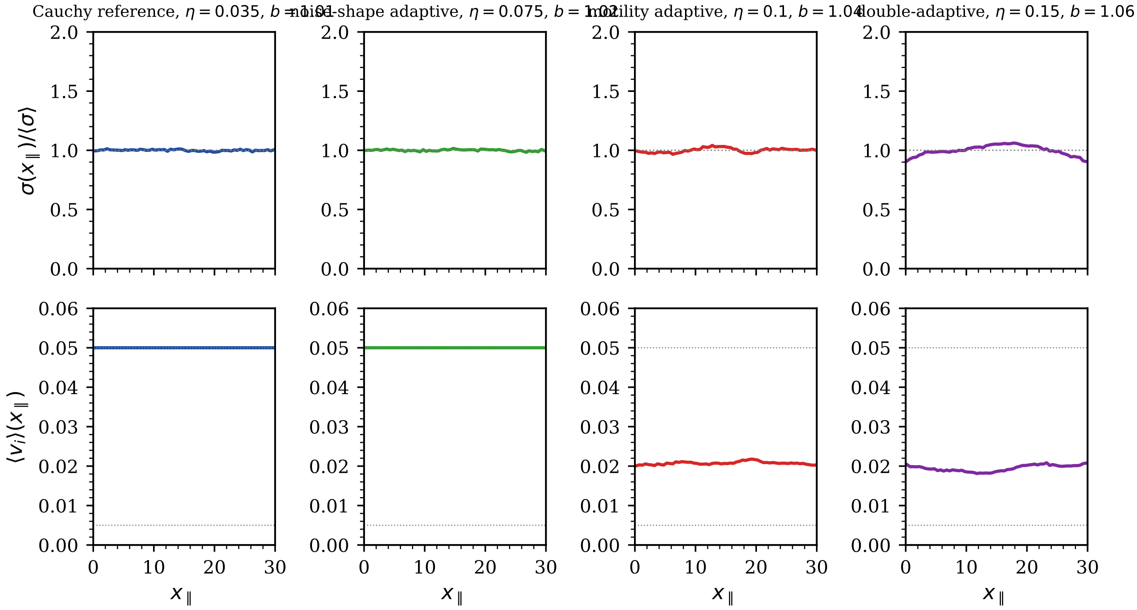

Figure 10. Time-averaged density (top row) and mean local speed (bottom row) along the instantaneous polar-flow axis $x_\parallel$ at $L = 30$, three seeds, 4000 measurement steps. The four columns are the four modes at their respective near-critical η values. The band-soliton index b is reported in each panel title.

All four profiles are flat to within 7 % ($b = 1.01, 1.02, 1.04, 1.06$ in the order Cauchy, noise-shape, motility, full). None supports a metric-Vicsek band: the dense phase generated by the motility channel either reorganises faster than its polar axis or is isotropic in the comoving frame. The full mode is marginally the most cohesive, consistent with the Gaussian rectification tightening the patch from within, but the increment is well below the soliton threshold.

3.2.5 Direct comparison with the original Vicsek model

The four heavy-tailed modes share a Cauchy backdrop and are directly comparable; the canonical Vicsek–Gaussian baseline sits outside. We recover it from Eqs. (1)–(6) with $v_{\min} = v_{\max} = 0.05$ and $\alpha_{\min} = \alpha_{\max} = 2$ on the same seven-size, three-seed protocol. The reference does not develop a critical $\chi_{\max}(L)$ scaling at our density: the peak stays in 1.1–1.4 across the seven sizes, slope $a_{\text{Vicsek}} = +0.01$ ($[-0.09, +0.06]$); the metric-Vicsek susceptibility is damped below the resolution of our lever. The spatial diagnostic separates more sharply: $s_{\text{sep}} = 1.24$ for Vicsek at $L = 64$, nearly identical to Cauchy and noise-shape (1.25), within the homogeneous-fluid range as Cates–Tailleur theory anticipates for a fixed-speed model. The full mode lands at 1.72. The Binder cumulants are similar across the five modes ($U_4 \in [0.3, 0.5]$).

3.3 Local diagnostics

3.3.1 Cluster-size distribution

The cluster-size distribution complements s_{sep}: a homogeneous fluid carries small fluctuations, a spinodal regime a power-law tail with 2D exponent near -2, a binodal regime one giant cluster coexisting with small ones. We coarse-grain on a 10×10 grid, threshold each cell at 1.5 times the median occupancy, and run periodic-aware connected-component labelling on the binary mask. A grid-free DBSCAN check ($\varepsilon = R_a$, four-point core) on the top-density quartile ranks the modes consistently and confirms the full mode carries the larger maximum cluster; we keep the grid-threshold counts here and ship DBSCAN with the code. Ten seeds at $L = 30$ separate the four modes (Figs. 11, 12).

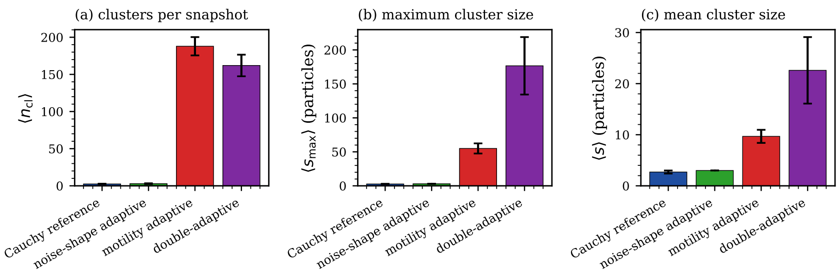

Figure 11. Cluster summary statistics per mode at $L = 30$, $N = 2000$, ten seeds. Panels (a)–(c) report the seed-averaged cluster count, maximum size and mean size, each with ± 1 SE bars. The two reference modes sit near the threshold floor; motility-only fragments the dense phase into many medium clusters; the double-adaptive model concentrates the same mass into a few large droplets ($\langle s_{\max} \rangle = 177$ vs 55).

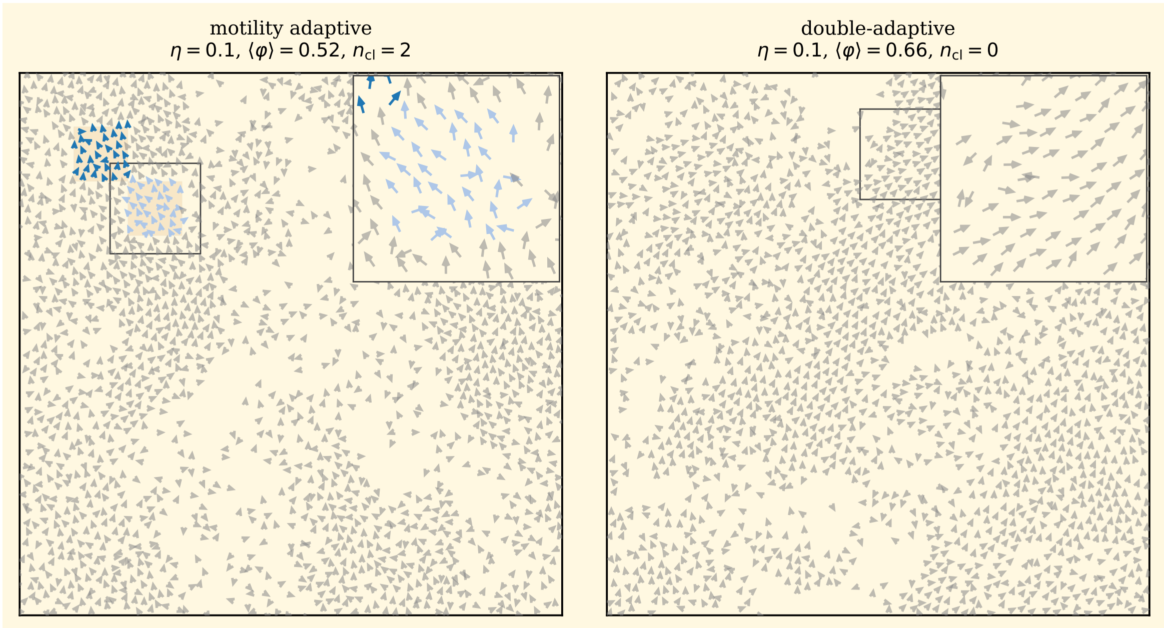

Figure 12. Real-space cluster identification at the near-critical point, $L = 30$, $N = 2000$, single seed, on a cream background. Particles in dense bins (above 1.5 times the median bin occupancy on a 10×10 grid) are drawn as arrows carrying the colour of the connected-component cluster they belong to; particles in dilute bins are grey. The dense bins are overlaid as a light ochre shading, and each panel carries a 5×5 zoom inset on its densest sub-region. Left: motility-only ablation, with many small-to-medium dense clusters scattered across the box. Right: double-adaptive model, with the same total dense area concentrated into one or two large droplets.

The Cauchy and noise-shape modes give only marginal density fluctuations (2.5 ± 0.4 and 2.9 ± 0.5 clusters per seed, max size near the threshold floor). Motility-only fragments the dense phase into 187.9 ± 12.3 clusters, max size 55.0 ± 7.5, mean 9.7 ± 1.3. The full mode gives 162.0 ± 14.5 clusters, max size 176.6 ± 42.3, mean 22.6 ± 6.5. The two motility-active modes differ on cluster count by -25.9 ± 22.6 ($z = -1.15$, not significant) and on max size by $+121.6 \pm 45.2$ ($z = +2.7$): the rectification fuses dense patches into a few large clusters at conserved total dense area (Fig. 12). Max size is the meaningful diagnostic here; the cluster count carries no clean signal.

3.3.2 Heading correlation $g(r)$ in dense and dilute regions

If the rectification really acts in the dense phase, its fingerprint should be visible independently of the cluster geometry. We measure the density-stratified heading correlation

$$g(r) = \langle \cos[\theta_i(t) - \theta_j(t)] \rangle_{|\mathbf{x}_i - \mathbf{x}_j| \in [r, r+\Delta r]} \tag{11}$$

for the four heavy-tailed modes at near-critical η, $L = 30$, three seeds with six snapshots per seed. The reference particle i is restricted to a density quartile of the local-density field $\rho_i = \#\{j : |\mathbf{x}_j - \mathbf{x}_i| < 1\}$; partner j ranges over the whole box (Fig. 13).

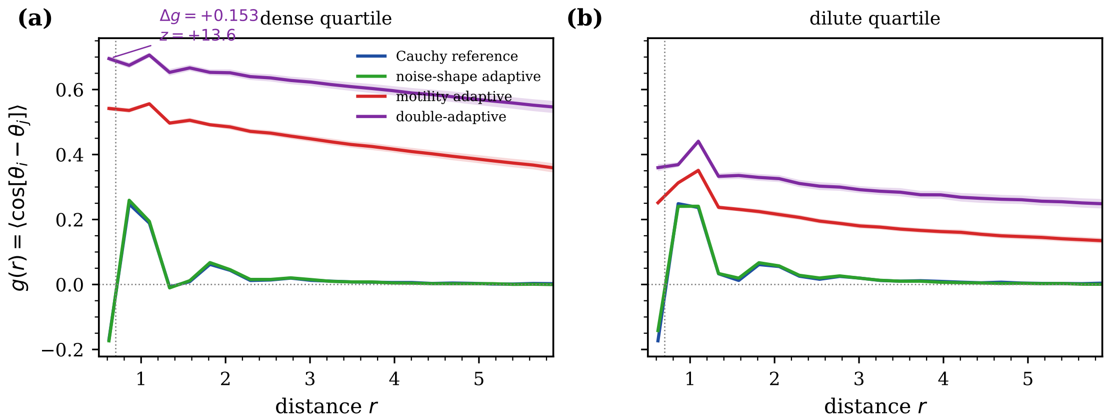

Figure 13. Heading correlation $g(r)$ in the dense (top quartile) and dilute (bottom quartile) regions of each heavy-tailed mode at the near-critical η, $L = 30$, $N = 2000$, three seeds, six snapshots per seed.

The dense-phase correlation splits the modes into two groups. The Cauchy and noise-shape modes reach $g(r \simeq 0.6) = -0.169 \pm 0.003$ and -0.173 ± 0.004 in the dense quartile, both negative: heavy-tailed bursts actively anti-align particles inside the dense patches. Motility-only reaches 0.542 ± 0.007, the full mode 0.695 ± 0.008, a paired $\Delta g = +0.153 \pm 0.011$ ($z = +14$). The dilute-phase correlations are essentially insensitive to the noise-shape adaptation: the asymmetry is the direct microscopic expression of the rectification, which acts where the shared sigmoid sends it and nowhere else.

The signature survives the FSS range. A ten-seed persistence test gives $\Delta g(r \simeq 0.6) = +0.150 \pm 0.010$ ($z = +15.7$) at $L = 64$, $+0.127 \pm 0.016$ ($z = +7.85$) at $L = 90$, and $+0.106 \pm 0.014$ ($z = +7.80$) at $L = 128$. Combined with $L = 30$, the dense-quartile gap plateaus at $\Delta g \in [+0.11, +0.15]$ across the four sizes, $z \geq 7$ throughout, with a mild narrowing from $+0.15$ at $L \leq 64$ to $+0.11$ at $L = 128$. The dense $g(r)$ is the diagnostic whose two-feedback signature survives all four sizes.

Extending the comparison to the full $r \in [0.5, 6]$ profile on the same ten-seed budget (Fig. 14), $\Delta g(r)$ is not a short-range fingerprint that decays with r. Its inner-range mean $r \in (0.5, 1.5)$ sits at $+0.150, +0.152, +0.131, +0.113$ for $L = 30, 64, 90, 128$, while its outer-range mean $r > 3$ climbs to $+0.181, +0.184, +0.157, +0.137$. The gap is as large beyond the alignment cutoff $R_a = 0.7$ as it is inside it, and slightly larger: the rectification imposes a long-range coherence on the dense phase that survives well past the metric interaction range.

3.3.3 Dynamic counterpart: heading autocorrelation $C(\tau)$

The dense $g(r)$ measures the static sharpness of local alignment, but says nothing about its persistence. The hydrodynamic appendix (Sec. A) predicts a rotational decorrelation rate $\Gamma(\rho, \eta) \simeq \eta^{\alpha(\rho)}$: η^2 inside the dense phase ($\alpha \to 2$), η in the dilute one ($\alpha \to 1$), a factor $\eta^{-1} \simeq 10$ at our working point. We measure the per-particle heading autocorrelation directly. For each mode at its near-critical η, $L = 30$, ten seeds, $n_{\text{meas}} = 20\,000$ steps, we tag every particle every 50 steps by quartile of the local count and compute $C_{\text{dense}}(\tau) = \langle\cos[\theta_i(t+\tau) - \theta_i(t)]\rangle_{i \in \text{dense}}$ on lags $\tau \in \{1, 2, 5, 10, 20, 50, 100, 200, 500, 1\,000, 2\,000, 5\,000\}$ (and similarly C_{dilute}).

The four modes order in the same way at every lag (Fig. 15a): Cauchy and noise-shape drop fastest, motility-only sits above, and the full model sits above motility-only by a nearly constant offset. The gap $\Delta C(\tau) = C_{\text{full}}(\tau) - C_{\text{motility}}(\tau)$ stays in the band $+0.13$ to $+0.24$ over $\tau \in [1, 2\,000]$ in the dense quartile, with bootstrap 95% CIs (ten seeds, $B = 10\,000$ resamples) excluding zero throughout; at $\tau = 5\,000$ the gap drops to $+0.18$ with CI $[+0.08, +0.27]$, still excluding zero. The

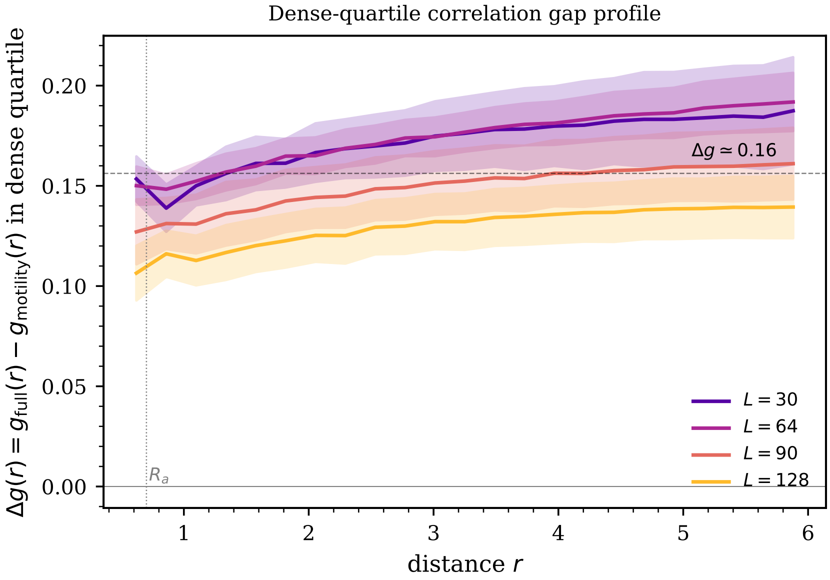

Figure 14. Dense-quartile correlation gap $\Delta g(r) = g_{\mathrm{full}}(r) - g_{\mathrm{motility}}(r)$ vs distance, ten seeds, $L \in \{30, 64, 90, 128\}$. Shaded bands: ± 1 seed-level SE. The gap plateaus near 0.11–0.18 across $r \in [0.5, 6]$ and rises mildly beyond $R_a = 0.7$.

within-seed estimator averages over $\sim 4 \times 10^7$ sample pairs per mode, so the between-seed SE is $\sim 10^{-3}$ and the t-statistic at nine d.o.f. saturates beyond a few sigmas; we therefore report bootstrap CIs. The dilute gap is a few percent and its CI brackets zero past $\tau \simeq 100$.

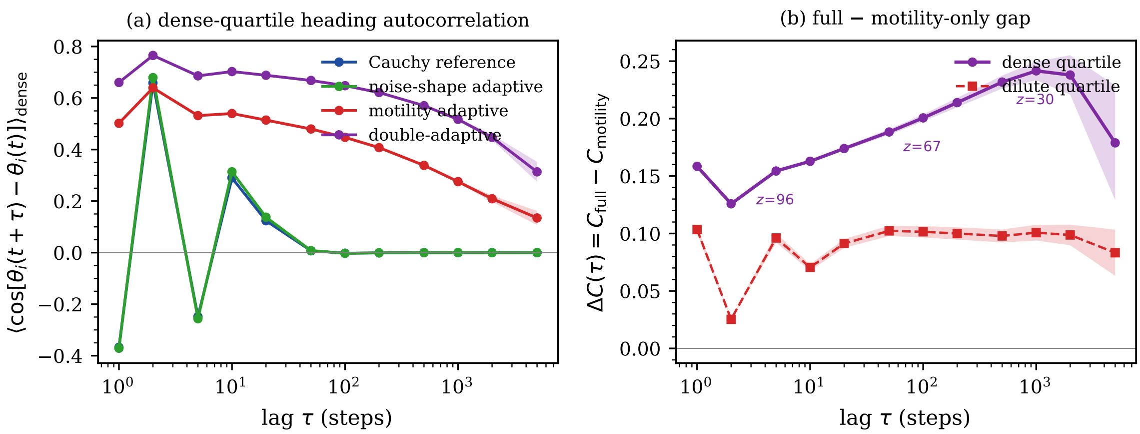

Figure 15. Density-stratified per-particle heading autocorrelation at $L = 30$, ten seeds, 20 000 steps. Panel (a): $C_{\mathrm{dense}}(\tau)$ per mode at near-critical η, ± 1 seed-level SE bands. Panel (b): gap $\Delta C(\tau) = C_{\mathrm{full}} - C_{\mathrm{motility}}$ in the dense and dilute quartiles, bootstrap 95% CIs as bands.

The dense-phase plateau matches the $\Gamma \sim \eta^{\alpha(\rho)}$ scaling of Sec. A.1: the gap is exactly the effect of switching α from 1 to 2 inside the dense phase built by the motility channel. The dense $g(r)$ captured the static counterpart in Fig. 14; $C(\tau)$ extends it to the single-particle dynamic at lags well beyond the polar-order correlation time.

3.3.4 Decoupled sigmoids: a four-way anatomy of the rectification

The shared sigmoid alone cannot say which ingredient does the work. We split it into two independent profiles and run four tests: the threshold plane (Sec. 3.3.4), the density dependence (Sec. 3.3.4), the alignment kernel (Sec. 3.3.4) and the temporal persistence of the patches

(Sec. 3.3.4). Together they narrow the rectification to a metric-specific, motility-driven consolidation of the instantaneous geometry that does not stabilise the patches in time.

We replace the shared sigmoid by two independent ones with parameters $(n_\star^v, s^v)$ and $(n_\star^\alpha, s^\alpha)$; setting both thresholds to 3 and both slopes to 2 recovers the shared model. Sweeping the quadrant $(n_\star^v, n_\star^\alpha) \in \{1, 2, 3, 5, 8\}^2$ at fixed $s^v = s^\alpha = 2$, ten seeds per cell at $L = 30$, $\eta = 0.15$, we record the seed-paired gap $\Delta g(r \simeq 0.6) = g_{\text{full}}^{\text{decoupled}} - g_{\text{motility}}$ (Fig. 16).

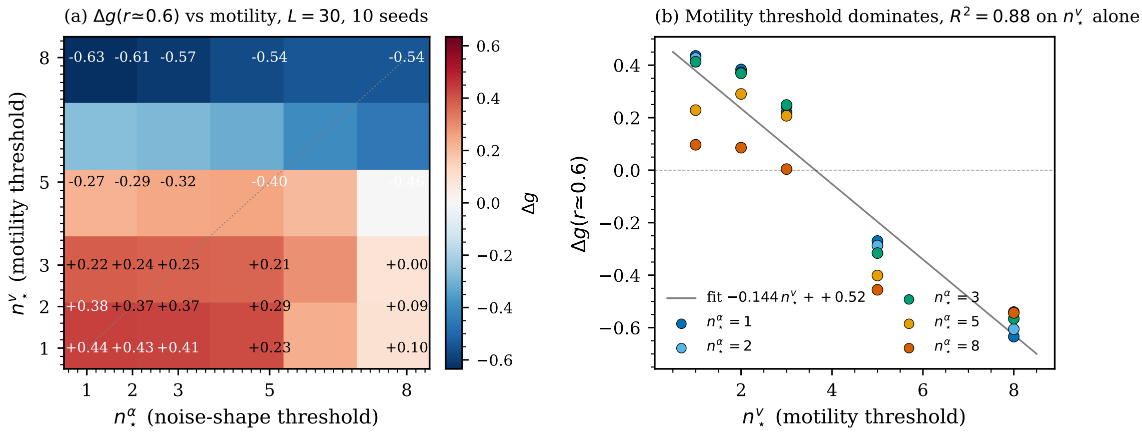

Figure 16. Decoupled-sigmoid threshold sweep at $L = 30$, ten seeds per cell. Panel (a): 25-cell heatmap of the paired gap $\Delta g(r \simeq 0.6) = g_{\text{full}}^{\text{decoupled}} - g_{\text{motility}}$; grey dashed line marks $n_\star^v = n_\star^\alpha$. Panel (b): same gap vs $n_\star^v$, colour-coded by $n_\star^\alpha$. The $n_\star^v$-only fit reaches $R^2 = 0.88$; $n_\star^\alpha$ alone, $R^2 = 0.04$.

The map splits the quadrant into two horizontal bands, separated not by the diagonal but by a critical motility threshold $n_\star^v \simeq 4$ — just above the typical alignment-zone occupancy $\sigma\pi(R_a^2 - R_r^2) \simeq 1.7$. For $n_\star^v \leq 3$ every row carries a positive gap, $+0.43 \pm 0.01$ at $(1, 1)$ down to $+0.10 \pm 0.01$ at $(1, 8)$; for $n_\star^v \geq 5$ every row is negative, -0.27 ± 0.01 at $(5, 1)$ down to -0.63 ± 0.01 at $(8, 1)$. The shared-sigmoid diagonal cuts straight through the stratification: $(3, 3)$ gives $+0.25$, $(5, 5)$ gives -0.40. The sign is set by $n_\star^v$ alone, not by the timing offset. Quantitatively, $\Delta g \simeq -0.144\, n_\star^v + 0.52$ captures $R^2 = 0.88$; the analogous fit on $n_\star^\alpha$ captures $R^2 = 0.04$; the timing-offset fit, $R^2 = 0.28$.

The mechanism is simple. At $n_\star^v = 3$, just above typical occupancy, dense-patch particles fire the motility channel and outside-patch particles saturate at $v_{\max}$. Push $n_\star^v$ to 5 or 8 and the motility channel almost never fires: the decoupled model collapses onto the noise-shape-only ablation, which has $g(r \simeq 0.6) = -0.17$ at $L = 30$ (Sec. 3.3.2); its gap with motility-only is large and negative, giving the lower band. Drop $n_\star^v$ back below 3 and the motility channel engages aggressively; dense patches grow and the noise-shape rectification adds coherence on top of them whether it triggers early or late, giving the upper band. The biological reading is direct: the density-triggered slowdown is a precondition. Heavy-tailed reorientations adapted to local crowding cannot build a flock on their own; they only sharpen the cohesion that the motility channel has already laid down.

We now ask how the critical motility threshold $n_\star^{v,\text{crit}}$ moves with the global density σ. The mean-density reading takes the relevant count to be $A\sigma = \pi(R_a^2 - R_r^2)\,\sigma$ and predicts $n_\star^{v,\text{crit}} \propto \sigma$ with slope $A \simeq +0.75$. We test that by sweeping $n_\star^v \in \{1, 2, 3, 4, 5, 6, 8\}$ at three additional densities $\sigma \in \{1.0, 1.5, 3.0\}$ (plus $\sigma = 2.22$), $n_\star^\alpha = 3$ fixed, ten seeds, and reading off the zero crossing of $\Delta g(n_\star^v) \simeq a(\sigma)\, n_\star^v + b(\sigma)$. Per-density fits all reach $R^2 \geq 0.86$ (Fig. 17) and

$$n_\star^{v,\text{crit}}(\sigma) = -0.73\,\sigma + 5.81, \qquad R^2 = 0.97. \tag{12}$$

The mean-density prediction is wrong in sign: $n_\star^{v,\text{crit}}$ falls with σ, the slope -0.73 the mirror image of $+0.75$.

What sets the sign is the dense-to-dilute density contrast. At low σ the dilute phase is nearly empty, the dense phase carries a large relative contrast, and its local neighbour count clears even a demanding $n_\star^v \simeq 5$. At high σ the dilute phase is itself crowded, the contrast weakens, and $n_\star^v$ has to drop for the rectification to remain selective. Extrapolating Eq. (12) to $\sigma \simeq 8$ predicts $n_\star^{v,\mathrm{crit}} \to 0$: once the two phases are indistinguishable, no threshold can produce a coherence gain.

A falsification check pushes the picture further. If $n_\star^{v,\mathrm{crit}}$ really tracked the dense-phase neighbour count, it should equal $\langle n_i \rangle_{\mathrm{dense}}$. We measure $\langle n_i \rangle_{\mathrm{dense}} = 2.57, 4.05, 4.85, 5.26$ at $\sigma = 1.0, 1.5, 2.22, 3.0$ (top-density quartile, $R = 1$, $L = 30$, 5 seeds, 12 snapshots), an *increasing* function of σ, opposite to $n_\star^{v,\mathrm{crit}}$. The ratio $n_\star^{v,\mathrm{crit}} / \langle n_i \rangle_{\mathrm{dense}}$ runs $2.08, 1.16, 0.85, 0.69$: the naive equality is refuted. A refined picture would make the threshold depend on how long the densest particles spend above it during consolidation; we leave that theory open.

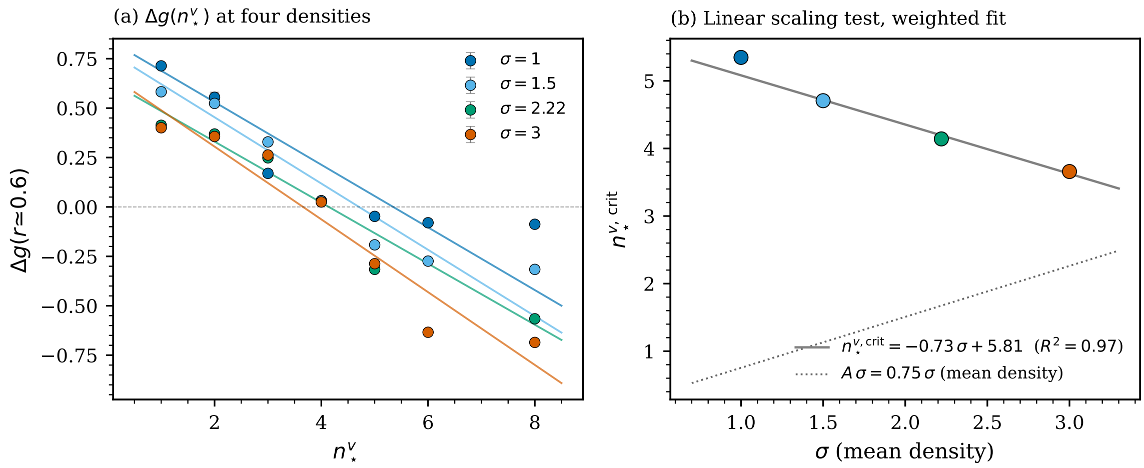

Figure 17. Density-scaling test. Panel (a): $\Delta g(r \simeq 0.6)$ vs $n_\star^v$ at four densities $\sigma \in \{1.0, 1.5, 2.22, 3.0\}$ (ten seeds, $n_\star^\alpha = 3$ fixed, $L = 30$), per-density weighted linear fits. Panel (b): critical threshold $n_\star^{v,\,\mathrm{crit}}(\sigma)$ extracted at the sign crossing. The weighted fit gives $n_\star^{v,\,\mathrm{crit}} = -0.73\,\sigma + 5.81$ ($R^2 = 0.97$); the dotted line is the mean-density prediction $A\sigma$ ($+0.75$).

Metric alignment conflates two roles: the sigmoid input n_i and the partner set are the same object. We pull the two apart with the topological variant of Ginelli and Chaté [2010], where each particle aligns with its k nearest non-repulsion neighbours while the sigmoid input stays the metric annulus count.

Re-running the 25-cell map at $L = 30$, $\sigma = 2.22$ with $k = 4$ gives the heatmap of Fig. 18b. The two-band structure softens. The linear fit on $n_\star^v$ alone now captures $R^2 = 0.68$ (was 0.88), while the $n_\star^\alpha$ fit climbs to $R^2 = 0.43$ (was 0.04). The bivariate fit $\Delta g \simeq -0.081\, n_\star^v - 0.069\, n_\star^\alpha + 0.73$ assigns the two thresholds nearly equal weight, against the $-0.144/-0.029$ split under metric alignment. The sign-flip region retreats: $(5, 1)$ and $(8, 1)$, which gave -0.27 and -0.63 under metric, now give $+0.31$ and $+0.21$; only the high-high corner stays negative.

The motility-threshold dominance is therefore a property of the metric kernel, not of the two-feedback dynamics. The reason is geometric: under metric alignment n_{ali} both feeds the sigmoid and selects the alignment partners, so pushing the threshold above typical occupancy switches off the slowdown and the neighbour-driven alignment together. Topological alignment breaks the identification: even with $n_\star^v$ high and the motility channel nearly silent, the $k = 4$ nearest neighbours hand each particle a baseline alignment that the noise-shape rectification can sharpen. Biologically, the model speaks sharpest to short-metric perceivers (swarming bacteria, insects in dense plumes) and less sharply to topological perceivers (starlings, schooling fish).

The droplets in the cluster diagnostic at $L = 30$ are larger in the full mode; are they longer-lived? We tracked the centroids of the connected dense components over 200 snapshots spaced 50 steps, matching them across snapshots by periodic-centroid distance (cutoff 4.0) and particle-count ratio (cutoff 2.0), ten seeds. The mean patch lifetime is 1.60 ± 0.02 snapshots in motility-only against 1.57 ± 0.02 in the full mode, a paired -0.037 ± 0.029 ($z = -1.30$); survival distributions

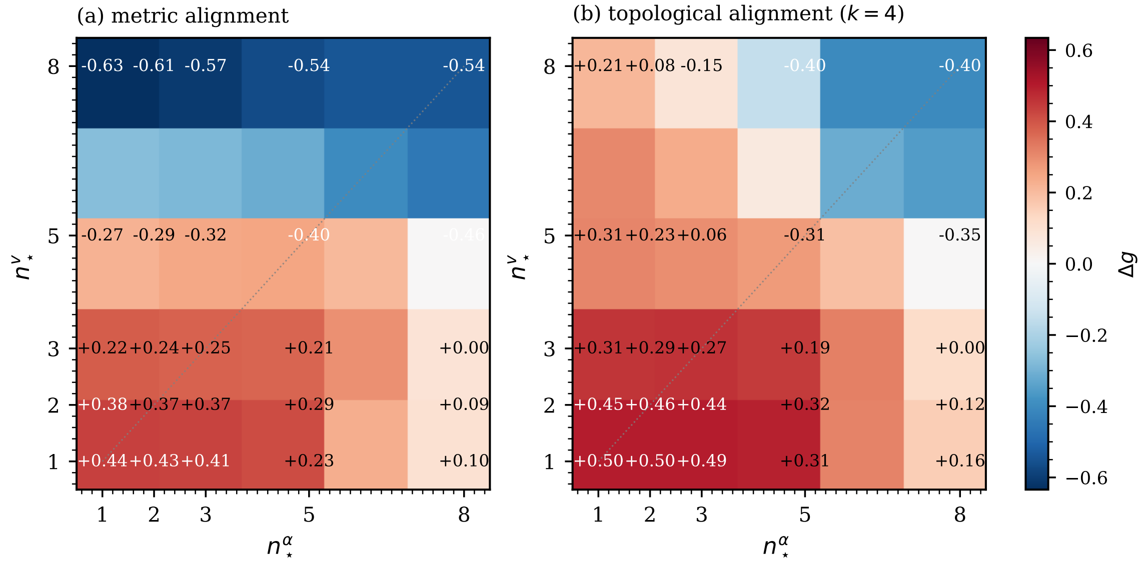

Figure 18. Alignment-kernel comparison of the 2D decoupled-sigmoid map at $L = 30$, $\sigma = 2.22$, ten seeds. Panel (a): metric (matched colour scale to Fig. 16). Panel (b): topological with $k = 4$. Grey dashed line: $n^v_\star = n^\alpha_\star$.

overlap below $\tau = 10$. The geometric consolidation does not buy temporal persistence: the noise-shape rectification reshapes which grid cells the dense particles occupy instantaneously while leaving cell turnover to the motility dynamics. It acts on geometry, not on persistence.

Together the four tests constrain the rectification sharply. It is a spatial consolidation that needs an aggressive motility threshold and a metric kernel to engage; under those conditions it works regardless of the noise-shape threshold, but it never extends patch lifetimes, and it switches off once the global density washes out the dense-to-dilute contrast. The mechanism is local in space and in time, and it piggybacks on the MIPS instability rather than driving the phase separation on its own.

4 Discussion

4.1 Mechanism and biological interpretation

What ties the two channels together is the shared sigmoid in n_i: speed and stability index move in lockstep. Inside a dense patch the sigmoid pushes $\alpha_i \to 2$, replacing the Cauchy bursts that would otherwise scramble a slow particle by Gaussian-like kicks. Each rectified kick no longer randomises the local alignment, and the noise-shape channel therefore acts as a fluctuation rectifier *inside* the slow region the motility channel carves out. The two adaptations cooperate where it matters and have no operating point at which they fight each other locally.

Four diagnostics read out the same mechanism on complementary axes. The FSS synergy on χ stays positive across the seven-size series, with the full mode scaling $\Delta a \simeq +0.6$ steeper than motility-only. The density-stratified dense $g(r)$ provides the static microscopic face ($\Delta g = +0.153$, $z = +14$ at $L = 30$, plateau $+0.11$ to $+0.15$ across $\{30, 64, 90, 128\}$); the dilute-quartile correlation is flat between the two modes, so the asymmetry is the direct fingerprint of the shared sigmoid acting only where it is meant to. The per-particle dense $C(\tau)$ extends the signature dynamically, holding $+0.13$ to $+0.24$ over four decades in lag. The cluster geometry adds the visual face: the maximum cluster size grows from 55 particles in motility-only to 177 in the full mode ($z = +2.7$). All four diagnostics are significant at $L = 30$.

The decoupled-sigmoid tests of Sec. 3.3.4 constrain the rectification further. It is a spatial-consolidation mechanism, not a temporal stabilisation (the patch-lifetime test returns $z = -1.3$);

it requires the metric kernel (the $n_\star^v$ dominance weakens from $R^2 = 0.88$ to 0.68 under topological alignment); it is controlled by the motility threshold alone under metric alignment; and it switches off altogether once the dense-to-dilute contrast washes out ($n_\star^{v,\mathrm{crit}} \to 0$ as $\sigma \to 8$).

A flock that adapts both its motility and its reorientation statistics to local crowding gets a spatially separated cohesive phase with reduced internal noise, a steeper susceptibility scaling than either channel alone, and a robust dense-phase heading correlation. The connection to locust marching [Buhl et al., 2006, Bazazi et al., 2008] and to swarming fish or starling flocks at sub-thermodynamic sizes is direct. The biological priority is unambiguous: a flock that slows down aggressively reaps the two-feedback cohesion at almost any noise-shape threshold; a flock with an inert motility channel sees the noise-shape adaptation damage dense-phase coherence. The slowdown is the precondition; the heavy-tail rectification is the sharpener.

4.2 Comparison to the literature

The motility channel inherits directly from Cates and Tailleur [2015]: our motility-only ablation reproduces $\partial \ln v / \partial \ln \rho < -1$ MIPS inside an aligning Couzin flock, with s_{sep} rising from $\simeq 1.2$ for the fixed-speed references to $\simeq 1.7$. The result is in the spirit of Bi et al. [2016]'s density-driven epithelial jamming; we keep the alignment interaction, so the dense phase is polar rather than amorphous. Within the metric-Vicsek lineage of Vicsek et al. [1995], Grégoire and Chaté [2004] and Chaté et al. [2008] established traveling bands and a weakly first-order transition; the band-soliton index $b \leq 1.06$ (Fig. 10) shows we do not reach the band regime, and $P(\langle \varphi \rangle)$ stays unimodal at $L = 30$ with $U_4 \simeq 0.65$ in both motility-only and full modes. The Toner–Tu programme [Toner and Tu, 1995, 1998, Toner, 2012] predicts the same continuous polar order but does not anticipate the dense-quartile gap of Fig. 13, since it treats the noise as Gaussian with density-independent variance. The $\Delta a \simeq +0.6$ gap between the full and motility-only slopes is a quantitative target for a density-dependent Toner–Tu closure.

The heavy-tail strand is the second axis. Reynolds [2018], Harris et al. [2012] and Ariel et al. [2015] document Lévy reorientations in foragers, T-cells and swarming bacteria. Our noise-shape ablation tempers the heavy-tail-as-driver picture: the Cauchy and noise-shape modes carry slopes statistically indistinguishable from zero, so heavy-tailed reorientations alone do not drive a critical $\chi_{\max}(L)$ scaling at our density — the alignment averaging in the dense phase washes the bursts out. The motility channel is necessary; heavy tails alone are not sufficient. What the noise-shape adaptation contributes is the extra cooperative $\Delta_n > 0$ on top of the motility-driven scaling. Couzin et al. [2002] treats the zonal parameters as global constants; we make the stability index a local field on the same kernel. Bechinger et al. [2016] and Marchetti et al. [2013] review and embed these feedbacks hydrodynamically; neither identifies the dense-phase Gaussian conversion because neither couples the noise distribution to the local density. Korobkova et al. [2004] and Dieterich et al. [2008] establish that bacterial reorientation statistics depend on the local environment, which makes the noise-shape feedback biologically sensible rather than phenomenologically convenient.

4.3 Limits and future work

The seven-size FSS places strong but not airtight bounds. The Δ_n trajectory is persistently positive with CI excluding zero at $n = 4, 5$; at $n = 6, 7$ the interval widens because the motility slope itself carries a wide bootstrap dispersion ($[+1.60, +3.04]$), but the point estimate stays positive. The non-monotonic drop from $\Delta_5 = +1.45$ to $\Delta_6 = +0.46$ tracks the slowing of the full $\chi_{\max}$ at $L = 90$, where the largest seed-level fluctuations sit; higher statistics at $L = 90$ would tell whether Δ_6 stabilises near $+0.5$ or drifts toward the asymptotic estimate $a_{\mathrm{full}} - a_{\mathrm{motility}} \simeq +0.6$. Controlled runs at $L = 180$ or $L = 256$ would resolve the residual Δ_n, tell whether the dense $g(r)$ signal plateaus near $+0.15$, and tell whether $P(\langle \varphi \rangle)$ eventually narrows.

Mapping $g(r)$ at intermediate distances $r = 2$ to $r = 5$ would clarify whether the rectification carries a characteristic decay length, in the spirit of Eliazar and Klafter [2003]. A two-knob sweep over $v_{\min}/v_{\max}$ and $\alpha_{\min}/\alpha_{\max}$ at fixed $(n_\star, s)$ would isolate the marginal contribution of each channel and bridge to the Vicsek limit. The absence of a traveling band leaves room for a different cohesive structure — a slowly-rotating dense droplet, perhaps — that a real-space analysis at $L > 30$ should reveal.

A topological alignment rule, shown to flip the heavy-tail trend in canonical Vicsek–Couzin, would be a one-line edit. A hydrodynamic theory recovering both the linear $\Delta g(n_\star^v)$ at fixed density and the negative slope $n_\star^{v,\mathrm{crit}}(\sigma) = -0.73\,\sigma + 5.81$ — attributable to the dense-to-dilute contrast rather than the mean density — would close the mechanistic argument. A Toner–Tu closure with $v_{\mathrm{eff}}(\rho)$ and a fractional $D_{\mathrm{rot}}(\rho) \propto \eta^{\alpha(\rho)}$ would put the mechanism on a quantitative footing; we plan to fit $\alpha_{\mathrm{eff}}(\rho)$ and $v_{\mathrm{eff}}(\rho)$ directly and benchmark the effective γ/ν against the metric Vicsek prediction. A movie utility shipped with the code writes MP4 sequences for each mode with frames coloured by v_i and an inset tracking $\langle\varphi\rangle(t)$ and $s_{\mathrm{sep}}(t)$, making the rectification visible in real time.

A Hydrodynamic description and the asymmetry of decoupled sigmoids

The microscopic model of Sec. 2 admits a coarse-grained description in terms of $\rho(\mathbf{x}, t)$ and $\mathbf{p}(\mathbf{x}, t) = \rho^{-1}\langle\vec{e}_i\rangle_x$. We sketch it at the level of physical arguments, the goal being to make the rectification mechanism analytically transparent.

A.1 Continuum fields and conservation

The instantaneous neighbour count in the continuum limit is

$$n_i \approx A\,\rho(\mathbf{x}_i), \qquad A \equiv \pi(R_a^2 - R_r^2), \tag{13}$$

with $A \simeq 0.75$ for $R_a = 0.7$ and $R_r = 0.5$. Substituting into Eqs. (4)–(6) replaces the per-particle adaptations by density fields

$$\sigma(\rho) = \frac{1}{1 + \exp\big[-s\,(A\rho - n_\star)\big]}, \tag{14}$$

$$v(\rho) = v_{\max} - \Delta v\,\sigma(\rho), \quad \Delta v \equiv v_{\max} - v_{\min}, \tag{15}$$

$$\alpha(\rho) = \alpha_{\min} + \Delta\alpha\,\sigma(\rho), \quad \Delta\alpha \equiv \alpha_{\max} - \alpha_{\min}. \tag{16}$$

Density conservation gives

$$\partial_t\rho + \nabla\cdot\big[\rho\,v(\rho)\,\mathbf{p}\big] = 0, \tag{17}$$

and to leading order in gradients and $|\mathbf{p}|$

$$\partial_t\mathbf{p} = -\Gamma(\rho,\eta)\,\mathbf{p} + \nu\,\nabla^2\mathbf{p} - \lambda\,(\mathbf{p}\cdot\nabla)\mathbf{p} + \cdots, \tag{18}$$

with ν, λ the standard Toner–Tu coefficients [Toner and Tu, 1995, Toner, 2012]. The non-trivial input is that Γ depends on ρ through $\alpha(\rho)$.

A.2 Effective rotational decorrelation

A symmetric α-stable kick $\xi \sim S_\alpha(c{=}\eta)$ has characteristic function $\langle e^{ik\xi}\rangle = \exp(-\eta^\alpha|k|^\alpha)$, so the single-step decorrelation of an aligned heading is $\langle\cos\xi\rangle = \exp(-\eta^\alpha)$ and, for small η,

$$\Gamma(\rho,\eta) = 1 - \langle\cos\xi\rangle \;\simeq\; \eta^{\alpha(\rho)}. \tag{19}$$

The decorrelation rate is η in the dilute limit (Cauchy) and η^2 in the dense limit (Gaussian); for $\eta < 1$ the ratio

$$\frac{\Gamma_{\mathrm{dense}}}{\Gamma_{\mathrm{dilute}}} = \eta^{\,\alpha_{\max}-\alpha_{\min}} = \eta \tag{20}$$

is the analytic counterpart of the rectification observed in the simulation: the noise-shape feedback turns the η-linear Cauchy decorrelation into the much weaker η^2 Gaussian one inside the dense phase.

A.3 MIPS instability and its rectification correction

A linear-stability analysis of the homogeneous polar state under a longitudinal density perturbation develops the Cates–Tailleur instability [Cates and Tailleur, 2015] when

$$\left.\frac{\partial \ln v(\rho)}{\partial \ln \rho}\right|_{\rho_0} < -1. \tag{21}$$

For the sigmoidal $v(\rho)$ of Eq. (15) this is met in a density window centred on $A\rho_0 \simeq n_\star$ of width $1/s$. The polar branch adds a term $\propto \Gamma(\rho_0, \eta) = \eta^{\alpha(\rho_0)}$. In the shared-sigmoid model $v(\rho)$ and $\alpha(\rho)$ vary together, so the instability window sits at the same density at which the rectification kicks in: the dense phase that nucleates is born already rectified.

A.4 Decoupled sigmoids: a sequential argument and its limits

Decoupling the sigmoid replaces Eq. (14) by two independent profiles with thresholds $n_\star^v$ and $n_\star^\alpha$, so MIPS is controlled by $\sigma_v(\rho)$ alone and the rectification by $\sigma_\alpha(\rho)$ alone. A sequential argument along the timing axis is instructive but incomplete.

For $n_\star^v < n_\star^\alpha$ (motility-first), MIPS triggers first at $A\rho_v \simeq n_\star^v$ with the noise still Cauchy; the dense phase grows to $A\rho_{\text{dense}} \gtrsim n_\star^\alpha$, at which Γ drops from η to η^2 *after* the patch has consolidated, and the dense-quartile $g(r)$ enhancement is predicted positive. For $n_\star^v > n_\star^\alpha$ (alpha-first), the rectification triggers first at $v \simeq v_{\max}$; fast Gaussian particles do not accumulate, but they take the Cauchy mean of their surrounding kicks at full force and are deflected harder than uniformly Cauchy particles would be, so the correlation is predicted to be damaged.

The sequential picture captures the diagonal ordering motility-first $\succ$ shared $\succ$ alpha-first, but misses the empirical map: Fig. 16 runs along horizontal lines of constant $n_\star^v$ ($R^2 = 0.88$ against $R^2 = 0.28$ for the timing-offset fit), and $(5, 8)$ stays firmly negative even though it is motility-first. The correct control variable is the engagement of the motility channel at typical local densities, not the timing offset; a Boltzmann coarse-graining of the full patch-formation dynamics would be needed to recover the $n_\star^v$-dominance quantitatively. We leave it open.

A.5 Toner–Tu interpretation

In Toner–Tu language with density-dependent coefficients, our model maps onto a flock with $v_{\text{eff}}(\rho)$ and a fractional rotational diffusion $D_{\text{rot}}(\rho) \propto \eta^{\alpha(\rho)}$. The fractional η-dependence is the imprint of the non-Gaussian angular noise. Reproducing the $n_\star^v$-dominance from Eqs. (17) and (18) requires identifying the critical motility threshold at which the dense-phase fraction collapses and propagating its effect through the noise-shape sector. The linear-stability sketch captures the diagonal two-band structure but neither the horizontal-band structure of the $(n_\star^v, n_\star^\alpha)$ map nor the σ-scaling law $n_\star^{v,\text{crit}}(\sigma) = -0.73\,\sigma + 5.81$ which runs opposite to the mean-density prediction. Whether the dense-phase local densities scale with σ as the MIPS contrast scales is the natural direction for a quantitative theory.

Code and data availability

The simulation code, all run drivers, the figure-generation scripts and the data are released as supplementary material. All runs reported here used Python 3.12.3, NumPy 2.4.4, SciPy 1.17.1, Numba 0.65.1 (LLVM 0.47.0) and Matplotlib 3.10.9 on Linux x86_64. The interaction loop is JIT-compiled with Numba's parallel reduction; per-particle random streams are derived from a single NumPy `Generator` with a per-run seed, so each `(seed, mode,` L`)` tuple is reproducible bit-for-bit.

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